

Classwork : Iteration 4th June 2026

Problem Statement 1: Water Tank Filling

A water tank is being filled with water at a constant rate of **10 liters per minute**. Initially, the tank contains **0 liters** of water.

Write a program that displays the amount of water in the tank after each minute and continues until the tank reaches **100 liters**.

Sample Output

Water Level: 10 liters
Water Level: 20 liters
Water Level: 30 liters
...
Water Level: 100 liters

Tank is full.

Problem Statement 2: Battery Charging

A mobile phone battery is charging. The battery level starts at **20%** and increases by **10%** every charging cycle.

Write a program that displays the battery percentage after each cycle and continues until the battery reaches **100%**.

Sample Output

Battery Level: 30%
Battery Level: 40%
Battery Level: 50%
...
Battery Level: 100%

Battery Fully Charged.

Problem Statement 3: Attendance Counter

A teacher is recording attendance as students enter the classroom. The class strength is **30 students**.

Write a program that counts the number of students entering the classroom and displays the attendance count until all 30 students are present.

Sample Output

Student Entered
Attendance Count: 1

Student Entered
Attendance Count: 2

...

Attendance Count: 30

All students are present.

Problem Statement 4: Attendance Tracker

A teacher is recording attendance as students enter the classroom. The class strength is **30 students**.

Write a program that inputs whether Student is Present or Absent. Display total number of student present as well as absent.

Sample Output

Student 1
Attendance : Present

Student 2
Attendance : Absent

...

No. of Students Present : 20
No. of Students Absent : 10

Problem Statement 5: PIN Verification

An ATM machine requires the user to enter the correct PIN to access their account. The valid PIN is **1234**.

Write a program that repeatedly asks the user to enter a PIN until the correct PIN is entered.

Sample Output

Enter PIN: 1111
Incorrect PIN. Try Again.

Enter PIN: 2222
Incorrect PIN. Try Again.

Enter PIN: 1234
Access Granted.

Problem Statement 6: Login System

A website allows users to log in using a password. The correct password is **admin123**.

Write a program that keeps asking the user to enter the password until the correct password is provided.

Sample Output

Enter Password: test
Invalid Password.

Enter Password: hello
Invalid Password.

Enter Password: admin123
Login Successful.

Problem Statement 7: Shopping Cart Bill

A customer is adding items to a shopping cart. The price of each item is entered one by one.

Write a program that continuously accepts item prices and calculates the total bill amount. The program should stop accepting prices when the user enters **0**.

Sample Output

Enter Item Price: 250
Enter Item Price: 100
Enter Item Price: 450
Enter Item Price: 0

Total Bill Amount: ₹800

Problem Statement 8: Guess the Number Game

A game has selected a secret number **7**. The player must keep guessing the number until the correct guess is made.

Write a program that repeatedly asks the user to guess the number and displays a success message when the correct number is entered.

Sample Output

Guess the Number: 3
Wrong Guess. Try Again.

Guess the Number: 5
Wrong Guess. Try Again.

Guess the Number: 7
Congratulations! You guessed the correct number.

Challenge Question (Real-World While Loop)

Online Exam Portal

A student must score at least **40 marks** to pass an online assessment. The system allows the student to reattempt the test until the passing score is achieved.

Problem Statement:

Write a program that accepts marks from the user and continues asking for marks until the entered score is **40 or more**. Display a congratulatory message once the student passes the assessment.

Sample Output

Enter Marks: 25
Result: Fail

Enter Marks: 32
Result: Fail

Enter Marks: 45
Result: Pass

Congratulations! You have cleared the assessment.